

እሺ። እሁን **Chapter 18 — Bank Integration & Core Banking Architecture** እንቀጥላለን። ይህ የEDGN በጣም ወሳኝ technical chapter ነው፤ ምክንያቱም **Guarantee** በEDGN ላይ ብቻ አይፈጠርም—የbank የcredit, collateral, approval እና actual financial execution ስርዓቶች ጋር መገናኘት አለበት።

EDGN — NATIONAL DIGITAL GUARANTEE NETWORK

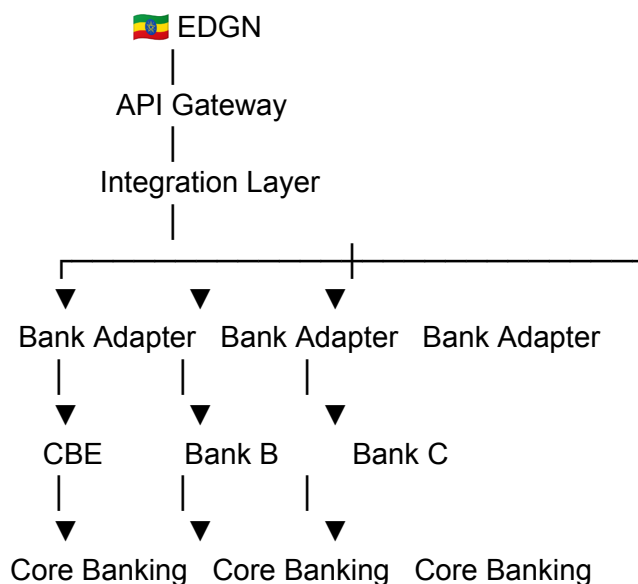
CHAPTER 18 — BANK INTEGRATION & CORE BANKING ARCHITECTURE

18.1 ዓላማ

የBank Integration Layer ዋና ዓላማ:

ከEDGN የሚመጣ **guarantee application** በ**secure, standardized, auditable API** ወደ bank ማስተላለፍ፤ bank የሚያደርገውን **customer verification, credit assessment, collateral management, approval, issuance, claim settlement** እና **release** ከEDGN ጋር ማመሳሰል ነው።

18.2 ዋና Architecture



| | |
Guarantee System / Credit / Collateral

ዋና መርህ: EDGN ለእያንዳንዱ bank የተለየ integration code ስcore ላይ እንዳይጠራቀም የ**Bank Adapter Layer** ይጠቀማል።

18.3 ሶስት ዋና Layers

Layer 1 — EDGN Core

Guarantee Engine
Registry
Workflow
Dispute
Notification
Audit

Layer 2 — Bank Integration

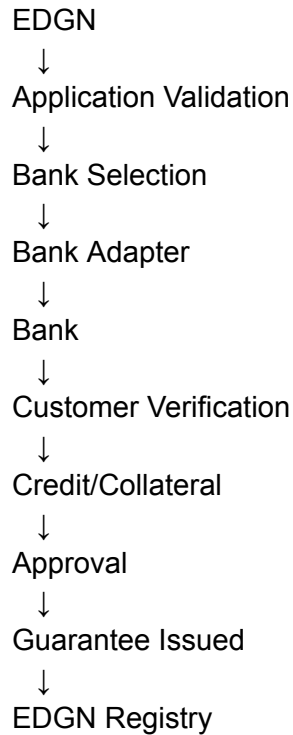
API Gateway
Bank Adapter
Message Queue
Event Processor
Transformation
Monitoring

Layer 3 — Bank Systems

Core Banking
Guarantee Module
Credit System
Collateral System
Customer/KYC
Accounting
Payment System

18.4 Guarantee Application Flow

Applicant
↓



18.5 Customer Verification

EDGN ወደ bank:

POST /bank/customers/verify

ሊልክ የሚችለው:

```
{
  "customerReference": "CUS-001",
  "nationalIdReference": "ID-REF",
  "bankCustomerReference": "optional"
}
```

Bank:

```
{
  "verified": true,
  "customerReference": "CUS-001",
  "status": "ACTIVE"
}
```

ትክክለኛ **API fields** በ**bank-specific contract** ይወሰናሉ ።

18.6 Account Eligibility

EDGN:

GET /bank/accounts/{accountReference}/eligibility

Bank:

Account Exists?

Active?

Eligible?

Restricted?

የbank policy የሚፈቅደው መረጃ ብቻ ይመለሳል።

18.7 Guarantee Application

POST /bank/guarantees/applications

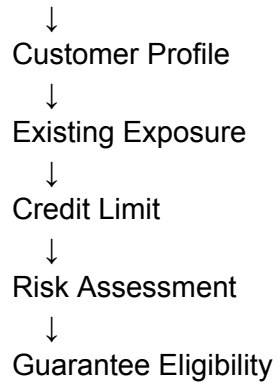
Payload:

```
{
  "applicationId": "GA-10001",
  "productType": "PERFORMANCE_GUARANTEE",
  "applicantReference": "CUS-001",
  "beneficiaryReference": "BEN-001",
  "amount": 5000000,
  "currency": "ETB",
  "expiryDate": "2027-01-31"
}
```

18.8 Bank Credit Assessment

ከEDGN ወደ bank ከሄደ በኋላ:

Application

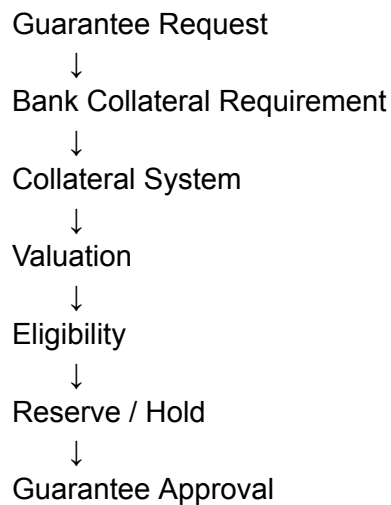


Credit decision የbank ነው።

EDGN ይህን አይተካም።

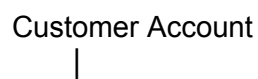
18.9 Collateral Architecture

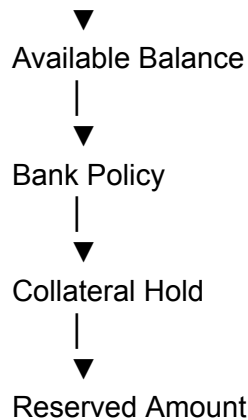
ይህ እጅግ ወሳኝ ነው።



18.10 Account-based Collateral

ለምሳሌ:





EDGN የaccount balance በራሱ እይቆጣጠርም።

Bank Core/Collateral System ነው የfund hold/reserve የሚያደርገው።

18.11 Collateral Reservation

POST /bank/guarantees/{id}/reserve

Example:

```
{
  "guaranteeld": "G-10001",
  "amount": 5000000,
  "currency": "ETB",
  "collateralReference": "COL-001"
}
```

Bank:

```
{
  "reserved": true,
  "reservationReference": "RES-10001"
}
```

18.12 Other Collateral Types

Bank እንደ policy ሊቀበል:

Cash
Account Balance
Fixed Deposit
Eligible Securities
Property
Other Approved Collateral

ነገር ግን **EDGN የcollateral valuation እና legal perfection** ስርዓት ለመተካት አይስራም።

18.13 Approval Workflow

Application
↓
Bank Officer
↓
Credit Review
↓
Collateral Review
↓
Maker
↓
Checker
↓
Approver
↓
Approved

ባንክ policy መሠረት approval levels ሊለያዩ ይችላሉ።

18.14 Maker-Checker

ባንክ guarantee issuance ላይ:

Maker
↓
Creates / Reviews
↓
Checker

↓
Validates
↓
Approver
↓
Issue

ይህ segregation of duties ለfinancial control ወሳኝ ነው።

18.15 Guarantee Issuance

Bank ከፀደቀ:

Bank
↓
Generate Guarantee Number
↓
Generate Official Guarantee Record
↓
Sign / Authorize
↓
EDGN

EDGN:

Receive
↓
Validate Signature/Reference
↓
Register
↓
Activate

18.16 Digital Signature

Guarantee document/data ላይ:

Bank Authorized Signatory
↓

Digital Signature



Guarantee



EDGN Registry

EDGN የsignature validity ለመፈተሽ የPKI/trust architecture ሊጠቀም ይችላል።

18.17 Digital Guarantee Document

BANK GUARANTEE	
Guarantee No	
Applicant	
Beneficiary	
Amount	
Currency	
Issue Date	
Expiry Date	
Purpose	
Status	
Digital Signature	
Verification QR	

18.18 EDGN Registry

Bank-issued guarantee ከተረጋገጠ:

Bank



EDGN



National Guarantee Registry

Registry:

Guarantee ID
Guarantee Number
Bank
Product
Applicant Reference
Beneficiary Reference
Amount
Issue Date
Expiry
Status
Version

18.19 Status Synchronization

Bank:

ISSUED
AMENDED
RENEWED
CLAIMED
CANCELLED
RELEASED

EDGN 事件:

Bank Event



Event Gateway



EDGN Event Processor



Registry Update

18.20 Event-driven Architecture

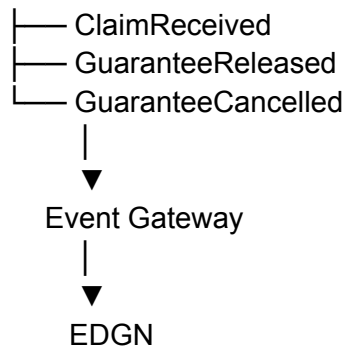
BANK



— GuaranteeIssued

— GuaranteeAmended

— GuaranteeRenewed



ይህ የreal-time/near-real-time synchronization እንዲኖር ይረዳል።

18.21 Webhook Architecture

Bank:

POST /edgn/events/guarantee-issued

EDGN:

Receive



Authenticate



Validate



Idempotency Check



Process



Acknowledge

18.22 Idempotency

እንደ event ሁለት ጊዜ ቢመጣ:

GuaranteeIssued

GuaranteeIssued

EDGN:

Event ID = EVT-001

Already Processed



Ignore Duplicate

ይህ በfinancial integrations ላይ በጣም ወሳኝ ነው።

18.23 Claim Integration

Beneficiary:

Claim



EDGN



Bank

Bank:

Claim Received



Validate



Review



Decision

18.24 Claim API

POST /bank/guarantees/{id}/claims

Response:

```
{
  "claimReference": "CLM-1001",
  "status": "RECEIVED"
}
```

18.25 Claim Settlement

Approved claim:

Bank



Settlement



Payment Infrastructure



Beneficiary



Confirmation



EDGN

EDGN:

Claim Status

Settlement Reference

Settlement Date

ይመዘግባል።

18.26 Release

Obligation ከተጠናቀቀ:

Beneficiary / Authorized Party



Release Request



EDGN



Bank



Guarantee Release



Collateral Release

18.27 Collateral Release

Guarantee Released



Bank Collateral System



Remove Hold



Available Balance Restored

ለምሳሌ:

Before:

Available = 10M

Reserved = 5M

After Release:

Available = 15M

Reserved = 0

ይህ ምሳሌ ብቻ ነው፤ actual accounting treatment በbank policy/core banking ይወሰናል።

18.28 Reconciliation

በየቀኑ/Inconfigured cycle:

EDGN Registry



Bank Guarantee System



Core Banking



Accounting

ይነጻጸራል።

18.29 Reconciliation Exceptions

ለምሳሌ:

EDGN = ACTIVE

Bank = RELEASED

ይህ:

 RECONCILIATION EXCEPTION

ይሆናል።

ሌሎች:

Amount mismatch

Status mismatch

Missing guarantee

Duplicate guarantee

Missing release

Settlement mismatch

18.30 Exception Management

Exception

↓

Case Created

↓

Bank Operations

↓

EDGN Operations

↓

Investigate

↓

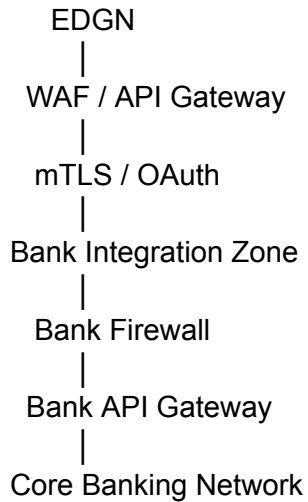
Correct

↓

Close

ማን እንደራታው መብት audit trail ይኖረዋል።

18.31 Security Architecture



Controls

- mTLS
 - OAuth2/service authentication
 - Certificate management
 - IP/network restrictions
 - RBAC
 - Encryption
 - API rate limiting
 - Request signing where required
 - Replay protection
 - Audit logging
 - Secrets management
-

18.32 HashiCorp Vault

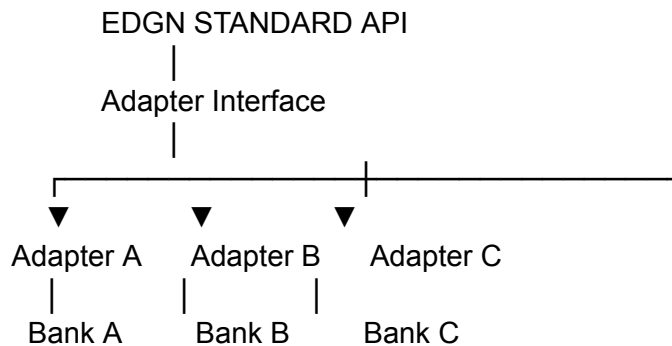
ከቀድሞው tech stack ጋር:

EDGN
↓
Vault
↓
API Credentials
Certificates
Encryption Keys
Secrets
↓

Bank Adapter

Credentials **source code** ውስጥ አይቀመጡም።

18.33 Bank Adapter Design



ለምሳሌ EDGN የሚጠራው:

POST /guarantees

Bank A ሌላ endpoint ሊኖረው ይችላል። Adapter ይተረጎመዋል።

18.34 Standardization

EDGN አንድ **Canonical Guarantee Data Model** ይኖረዋል።

Guarantee

- ├── Identity
- ├── Applicant
- ├── Beneficiary
- ├── Bank
- ├── Product
- ├── Amount
- ├── Currency
- ├── Dates
- ├── Status
- ├── References
- └── Documents

Bank-specific format nadapter 8492.6::

18.35 Core API Set

Customer

POST /customers/verify

Account

GET /accounts/{id}/eligibility

Guarantee

POST /guarantees

GET /guarantees/{id}

Collateral

POST /guarantees/{id}/reserve

GET /guarantees/{id}/collateral

Claim

POST /guarantees/{id}/claim

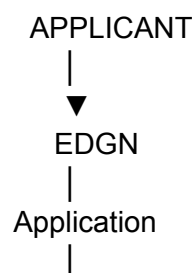
Release

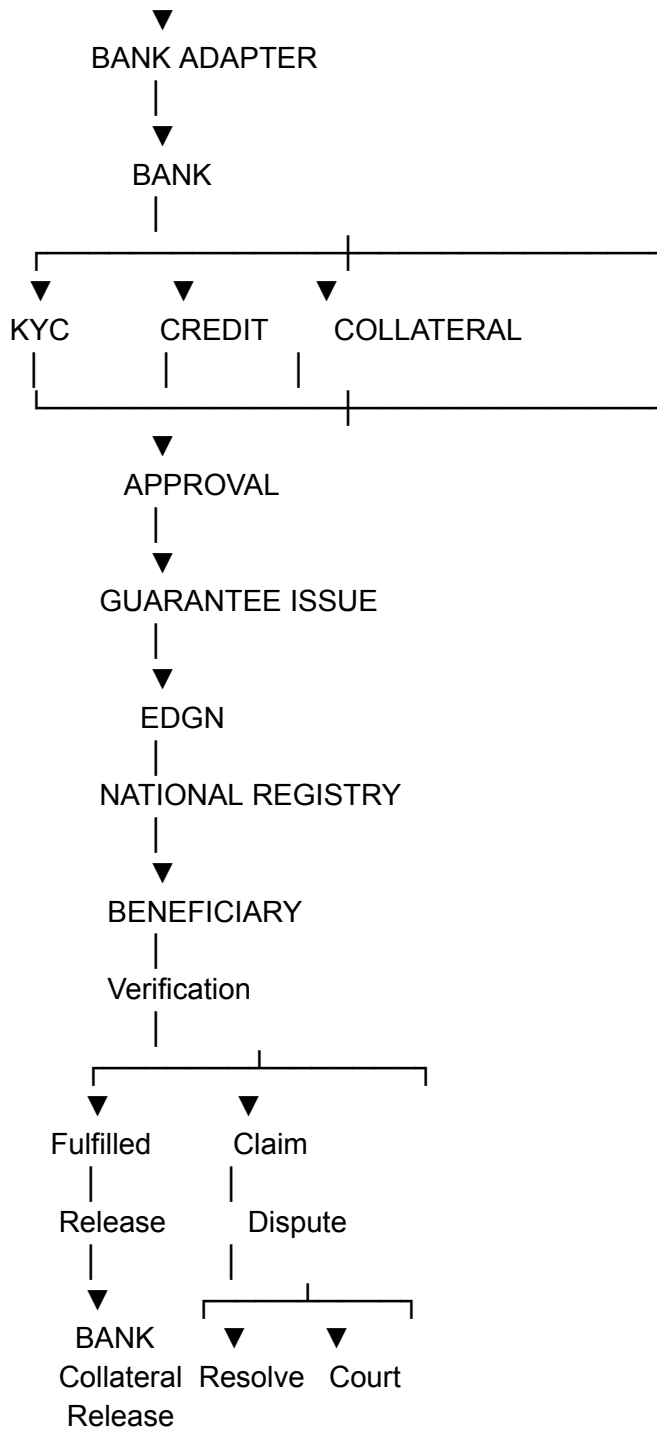
POST /guarantees/{id}/release

Transaction

GET /transactions/{id}

18.36 Complete Bank Flow





18.37 Bank Integration Monitoring

EDGN Admin:

BANK INTEGRATION MONITOR

Bank A  UP
Bank B  UP
Bank C  DEGRADED
Bank D  DOWN

Metrics:

API Success Rate

Latency

Timeouts

Failed Requests

Pending Requests

Event Backlog

Reconciliation Exceptions

18.38 Failure Handling

Bank API ክልራ፡

EDGN

↓

Request

↓

Timeout

↓

Retry Policy

↓

Retry

↓

Still Failed?

↓

Queue

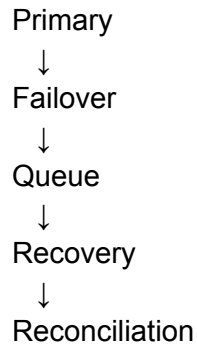
↓

Manual/Automated Recovery

Financial transaction ላይ **blind retry** አይደረግም። Idempotency/reference verification ይጠበቃል።

18.39 Disaster Recovery

Bank integration ለንዳይቋረጥ:



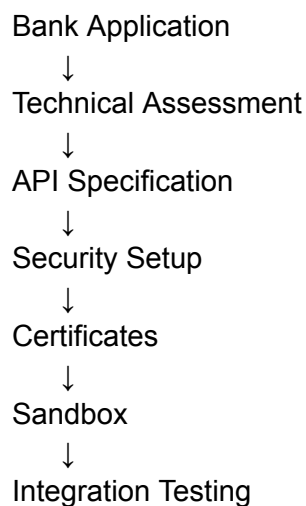
EDGN ለcritical services:

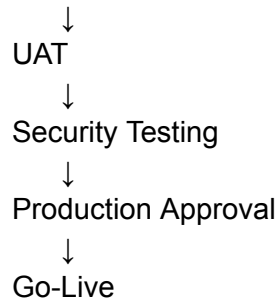
- backup
- replication
- monitoring
- recovery procedures
- audit preservation

ሊኖረው ይገባል።

18.40 Bank Onboarding Process

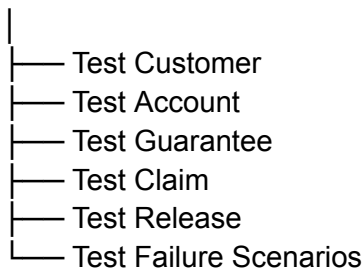
አዲስ bank EDGN ላይ ሲገናኝ:





18.41 Sandbox Environment

EDGN SANDBOX



Production ላይ ከመግባት በፊት:

Functional Test
Security Test
Performance Test
Failure Test
Reconciliation Test

መሟላት አለበት።

18.42 Bank Integration Governance

እያንዳንዱ bank:

Integration Agreement
API Contract
Security Requirements
SLA
Incident Process

Data Sharing Rules
Audit Requirements
Change Management

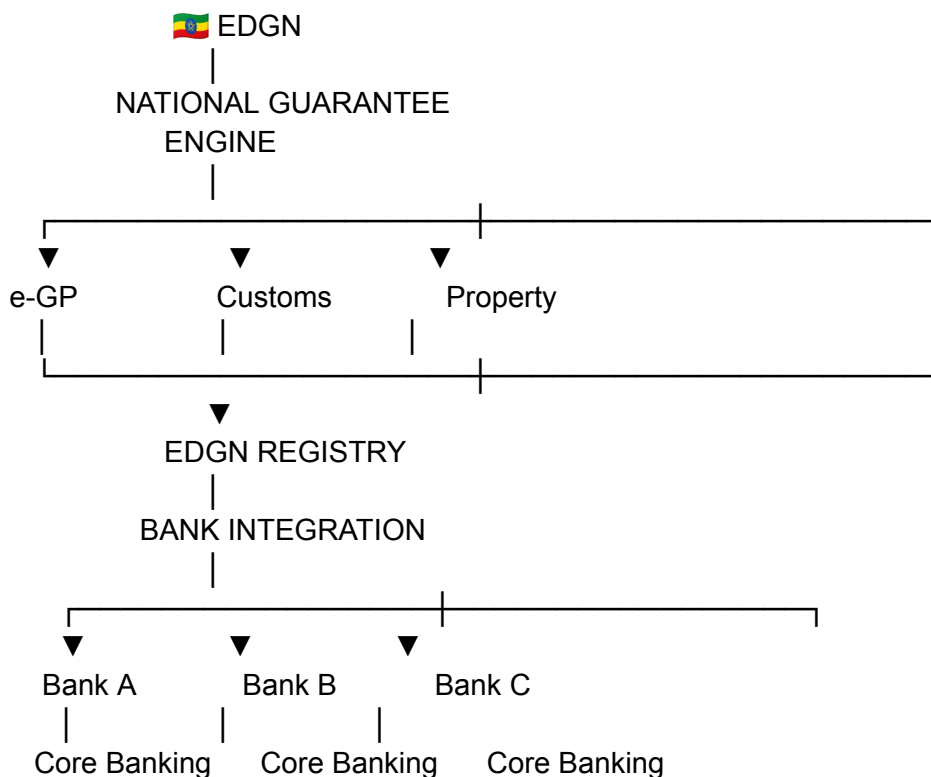
ይኖረዋል።

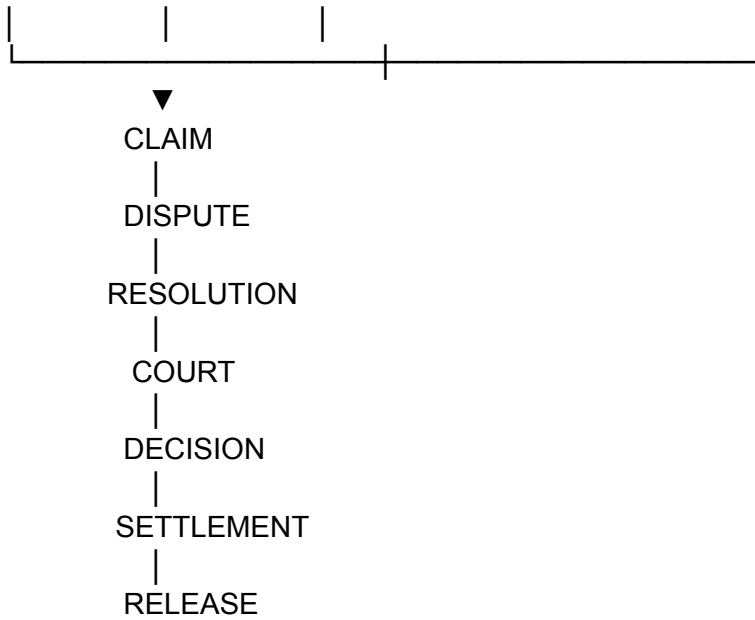
18.43 ትልቁ የTechnical Architecture መርህ

EDGN does not become the bank. EDGN becomes the standardized national digital guarantee orchestration and verification layer connecting authorized banks with government, businesses, beneficiaries and other approved ecosystems.

ይህን መርህ በproposal ውስጥ እንደ Core Architecture Principle እናስገባለን።

18.44 ከChapter 14–18 ያለን ትልቁ Architecture





✓ MASTER PROPOSAL STATUS

Chapter	Status
1–17	✓
18. Bank Integration & Core Banking Architecture	✓

▶▶ 🔑 — CHAPTER 19



NATIONAL GUARANTEE REGISTRY, DIGITAL VERIFICATION & ANTI-FRAUD ARCHITECTURE

በዚያ እንገነባለን፦

National Guarantee ID → QR Verification → Digital Signature → Tamper Detection → Duplicate Guarantee Detection → Fraud/Risk Engine → Blacklist/Watchlist → Audit Trail → Regulator Dashboard → Real-time Verification API

ይህ Chapter 19 ነው የEDGNን ከተለመደ “**guarantee management system**” ወደ እውነተኛ **National Digital Guarantee Infrastructure** የሚያደርሰው።